

- 27.** S_1 and S_2 are two circles on a plane with radii 4 cm, and 2 cm, respectively and the distance between their centres is 3 cm. Which one of the following statements is true?

(a) S_2 lies entirely within the circle S_1
 (b) S_1 and S_2 touch each other internally
 (c) S_1 and S_2 touch each other externally
 (d) S_1 and S_2 intersect in two distinct points

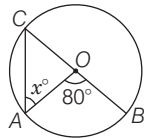
- 28.** ACB is a tangent to a circle at C , CD and CE are chords such that $\angle ACE > \angle ACD$. If $\angle ACD = \angle BCE = 50^\circ$, then

(a) $CD = CE$
 (b) ED is not parallel to AB
 (c) ED passes through the centre of the circle
 (d) $\triangle CDE$ is a right angled triangle

- 29.** A bicycle is running straight towards North. What is the locus of the centre of the front wheel of the bicycle whose diameter is d ?

(a) A line parallel to the path of the wheel of the bicycle at a height d cm
 (b) A line parallel to the path of the wheel of the bicycle at a height $d/2$ cm
 (c) A circle of radius $d/2$ cm
 (d) A circle of radius d cm

- 30.** If 'O' is the centre of circle, then x is equal to



(a) 80° (b) 60° (c) 40° (d) 20°

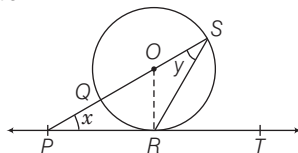
- 31.** If two equal circles touch each other externally, the common tangent divides the line of centres in the ratio

(a) 1 : 1 (b) 2 : 1 (c) 1 : 2 (d) 3 : 2

- 32.** With the vertices of a $\triangle ABC$ as centre three circles are described, each touching the other two circle externally. If the sides of the triangle are 9 cm, 7 cm and 6 cm. Then, the radius of the circle are

(a) 4, 5, 2 (b) 4, 5, 6
 (c) 3, 2, 3 (d) All equal to 3 cm

- 33.** In the given figure PT touches the circle with centre O at R . Diameter SQ when produced meet PT at P . If $\angle SPR = x$ and $\angle QSR = y$, then $x + 2y$ is equal to

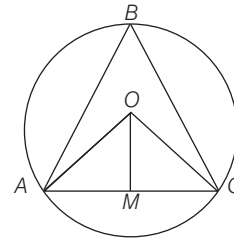


(a) 180° (b) 90°
 (c) 135° (d) None of these

- 34.** What is the locus of centres of circles which touch a given line at a given point?

(a) A line perpendicular to the given line, passing through the given point
 (b) A line parallel to the given line
 (c) A circle tangent to the given line at the given point
 (d) A closed curve other than a circle

- 35.** In the given figure, O is the centre of the circle, $OA = 3$ cm, $AC = 3$ cm and OM is perpendicular to AC . What is $\angle ABC$ equal to?



(a) 60° (b) 45°
 (c) 30° (d) None of these

- 36.** The distance between the centres of two circles having radii 4.5 cm and 3.5 cm, respectively is 10 cm. What is the length of the transverse common tangent of these circles?

(a) 8 cm (b) 7 cm
 (c) 6 cm (d) None of these

- 37.** ABC is an equilateral triangle inscribed in a circle with $AB = 5$ cm. Let the bisector of the angle, A meet BC in X and the circle in Y . What is the value of $AX \cdot AY$?

(a) 16 cm^2 (b) 20 cm^2 (c) 25 cm^2 (d) 30 cm^2

- 38.** Two unequal circles are touching each other externally at P , APB and CPD are two secants cutting the circles at A, B, C and D . Which one of the following is correct?

(a) $ACBD$ is parallelogram (b) $ACBD$ is a trapezium
 (c) $ACBD$ is a rhombus (d) None of these

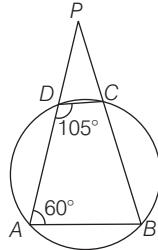
- 39.** $ABCD$ is a quadrilateral, the sides of which touch a circle. Which one of the following is correct?

(a) $AB + AD = CB + CD$ (b) $AB : CD = AD : BC$
 (c) $AB + CD = AD + BC$ (d) $AB : AD = CB : CD$

- 40.** Let PAB be a secant to a circle intersecting at points A and B and PC is a tangent. Which one of the following is correct?

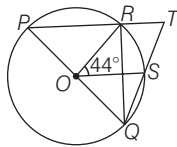
(a) The area of rectangle with PA, PB as sides is equal to the area of square with PC as sides
 (b) The area of rectangle with PA, PC as sides is equal to the area of square with PB as sides
 (c) The area of rectangular with PC, PB as sides is equal to the area of square with PA as side
 (d) The perimeter of rectangle with PA, PB as sides is equal to the perimeter of square with PC as side

41. In the given figure, if $\angle BAD = 60^\circ$, $\angle ADC = 105^\circ$, then what is $\angle DPC$ equal to?



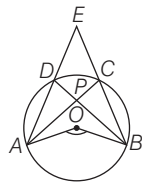
- (a) 40° (b) 45° (c) 50° (d) 60°

42. In the given figure, PQ is a diameter of the circle whose centre is at O . If $\angle ROS = 44^\circ$ and OR is a bisector of $\angle PRQ$, then what is the value of $\angle RTS$?



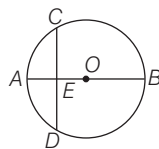
- (a) 46° (b) 64°
(c) 69° (d) None of these

43. In the given figure, O is the centre of the circle. AC and BD intersect at P . If $\angle AOB = 100^\circ$ and $\angle DAP = 30^\circ$, what is $\angle APB$?



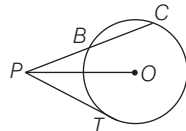
- (a) 77° (b) 80° (c) 85° (d) 90°

44. In the given figure, AB is a diameter of a circle and CD is perpendicular to AB , if $AB = 10$ cm and $AE = 2$ cm, then what is the length of ED ?



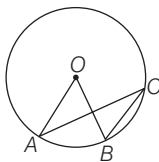
- (a) 5 cm (b) 4 cm (c) $\sqrt{10}$ cm (d) $\sqrt{20}$ cm

45. In the given figure, PT is a tangent to a circle of radius 6 cm. If P is at a distance of 10 cm from the centre O and $PB = 5$ cm, then what is the length of the chord BC ?



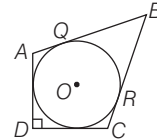
- (a) 7.8 cm (b) 8 cm (c) 8.4 cm (d) 9 cm

46. In the given figure, $\angle AOB = 46^\circ$, AC and OB intersect each other at right angles. What is the measure of $\angle OBC$? (O is the centre of the circle.)



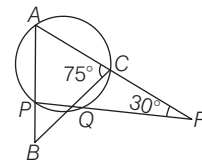
- (a) 44° (b) 46° (c) 67° (d) 78.5°

47. In the given figure, a circle is inscribed in a quadrilateral $ABCD$. Given that, $BC = 38$ cm, $QB = 27$ cm, $DC = 25$ cm and AD is perpendicular to DC . What is the radius of the circle?



- (a) 11 cm (b) 14 cm (c) 15 cm (d) 16 cm

48. In the figure given below, what is $\angle CBA$?



- (a) 30° (b) 45° (c) 50° (d) 60°

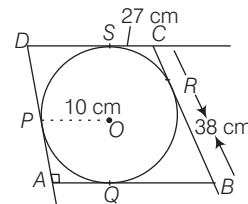
49. PQ is a common chord of two circles. APB is a secant line joining points A and B on the two circles. Two tangents AC and BC are drawn. If $\angle ACB = 45^\circ$, then what is $\angle AQB$ equal to?

- (a) 75° (b) 90° (c) 120° (d) 135°

50. $ABCD$ is concyclic quadrilateral. The tangents at A and C intersect each other at P . If $\angle ABC = 100^\circ$, then what is $\angle APC$ equal to?

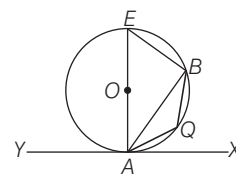
- (a) 10° (b) 20° (c) 30° (d) 40°

51. In the given figure, quad. $ABCD$ is circumscribed, touching the circle at P, Q, R and S such that $\angle DAB = 90^\circ$. If $CS = 27$ cm and $CB = 38$ cm and radius of the circle is 10 cm, then $AB = ?$



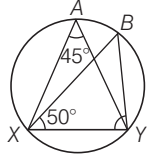
- (a) 28 cm (b) 21 cm (c) 19 cm (d) 17 cm

52. In the figure given below, YAX is a tangent to the circle with centre O . If $\angle BAX = 70^\circ$ and $\angle BAQ = 40^\circ$, then what is $\angle ABQ$ equal to?



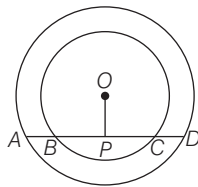
- (a) 20° (b) 30°
(c) 35° (d) 40°

53. In the given figure, what is $\angle BYX$ equal to?



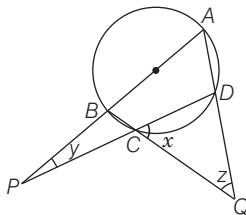
- (a) 85° (b) 50° (c) 45° (d) 90°

54. In the given figure, AD is a straight line, OP perpendicular to AD and O is the centre of both circles. If $OA = 20$ cm, $OB = 15$ cm and $OP = 12$ cm, what is AB equal to?



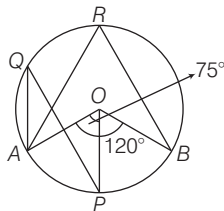
- (a) 7 cm (b) 8 cm (c) 10 cm (d) 12 cm

55. In the given figure, if $\frac{x}{3} = \frac{y}{4} = \frac{z}{5}$, where $\angle DCQ = x$, $\angle BPC = y$ and $\angle DQC = z$, then what are the values of x, y and z , respectively?



- (a) $33^\circ, 44^\circ$ and 55° (b) $36^\circ, 48^\circ$ and 60°
(c) $39^\circ, 52^\circ$ and 65° (d) $42^\circ, 56^\circ$ and 70°

56. In the given figure, if $\angle AOP = 75^\circ$ and $\angle AOB = 120^\circ$, then what is $\angle AQP$?



- (a) 45° (b) 37.5° (c) 30° (d) 22.5°

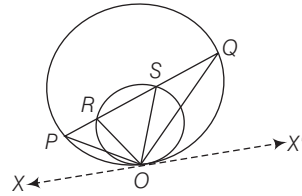
57. A circle of radius 5 cm has chord RS at a distance of 3 units from it. Chord PQ intersects with chord RS at T such that $TS = 1/3$ of RT . Find minimum value of PQ .

- (a) $6\sqrt{3}$ (b) $4\sqrt{3}$ (c) $8\sqrt{3}$ (d) $2\sqrt{3}$

58. Two mutually perpendicular chords AB and CD intersect at P . $AP = 4$ cm, $PB = 6$ cm, $CP = 3$ cm. Find radius of the circle.

- (a) $\frac{5\sqrt{5}}{2}$ cm (b) $\frac{3\sqrt{5}}{4}$ cm (c) $\frac{6\sqrt{5}}{2}$ cm (d) $\frac{2\sqrt{5}}{3}$ cm

- 59.



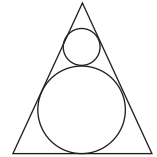
In the above figure which of the following holds good?

- (a) $\angle SOQ = \angle ROP$ (b) $2\angle ROP = \angle SOR$
(c) $\angle POR = \angle ASO$ (d) $\angle QOX' = \angle SOR + \angle ROP$

60. Semi-circle C_1 is drawn with a line segment PQ as its diameter with centre at R . Semicircles C_2 and C_3 are drawn with PR and QR as diameters respectively, both C_2 and C_3 lying inside C_1 . A full circle C_4 is drawn in such a way that it is tangent to all the three semicircles C_1, C_2 and C_3 , C_4 lies inside C_1 and outside both C_2 and C_3 . The radius of C_4 is

- (a) $\frac{1}{3}PQ$ (b) $\frac{1}{6}PQ$ (c) $\frac{1}{\sqrt{2}}PQ$ (d) $\frac{1}{4}PQ$

61. Two circles are placed in an equilateral triangle as shown in the figure. What is the ratio of the area of the smaller circle to that of the equilateral triangle?



- (a) $\pi : 36\sqrt{3}$ (b) $\pi : 18\sqrt{3}$ (c) $\pi : 27\sqrt{3}$ (d) $\pi : 42\sqrt{3}$

62. Consider the following statements :

- I. The opposite angles of a cyclic quadrilateral are supplementary.
II. Angle subtended by an arc at the centre is double the angle subtended by it at any point on the remaining part of the circle.

Which one of the following is/are correct in respect of the above statements?

- (a) Statement I $\Rightarrow$ Statement II
(b) Statement II $\Rightarrow$ Statement I
(c) Statement I $\Leftrightarrow$ Statement II
(d) Neither Statement I $\Rightarrow$ Statement II nor Statement II $\Rightarrow$ Statement I

63. Consider the following statements :

- I. Let P be a point on a straight line L . Let Q, R and S be the points on the same plane containing the line L such that PQ, PR and PS are perpendicular to L . Then, there exists no triangle with vertices Q, R, S .
II. Let C be a circle passing through three distinct points D, E and F such that the tangent at D to the circle C is parallel to EF . Then, DEF is an isosceles triangle.

Which of the statement(s) given above is/are correct?

- (a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

- 64.** The incircle of a $\triangle ABC$ touches the sides AB, BC and AC at the points P, Q, R , respectively, then which of the following statement is/are correct?

I. $AP + BQ + CR = PB + QC + RA$
 II. $AP + BQ + CR = \frac{1}{2}$ (perimeter of $\triangle ABC$)

III. $AP + BQ + CR = 3(AB + BC + CA)$

Select the correct answer using the codes given below

- (a) I, II and III are correct (b) Only I is correct
 (c) II and III are correct (d) I and II are correct

- 65.** Points X and Y are two different points on a circle. Point M is located so that line segment XM and line segment YM have equal length. Which of the following could be true?

I. M is the centre of the circle.
 II. M is on arc XY III. M is outside of the circle.
 (a) Only 1 (b) Only II (c) I and II (d) I, II and III

- 66.** Which of the following statement(s) are correct?

- I. A circle is the locus of points equidistant from a fixed point.
 II. The locus of centre of circle rolling on the circumference of fixed circle is a circle concentric with fixed circle.
 III. The locus of the points which are equidistant from three distinct points on a line, is a line parallel to the given line.
 IV. If the bisectors of $\angle B$ and $\angle C$ of a quadrilateral $ABCD$ intersect in P , then P is equidistant from AB and CD .

Select the correct answer using the codes given below

- (a) All are correct (b) Only IV
 (c) Both I and II (d) I, II and IV

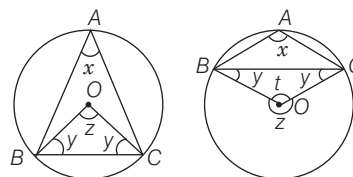
- 67.** A square and a circle intersect at more than one point. Consider the following statements:

- I If atleast two of the intersection points are on the vertices of the square then the square is cyclic.
 II If there are exactly four intersection points then the square has less area than the circle.
 III If there are exactly four intersection point then the diameter of circle is equal to the diagonal of square.
 IV If atleast two intersection points are on the vertices of square. Then it may be possible to have the sides of square equal to the diameter of circle.

Which of the statement (s) given above is/are correct ?

- (a) Only I (b) Only IV
 (c) Both I and II (d) II, III and IV

Directions (Q. Nos. 68-69) BC is a chord with centre O . A is a point on an arc BC as shown in the figure below.



- 68.** What is the value of $\angle BAC + \angle OBC$ if A is the point on the major arc?

- (a) 110° (b) 70° (c) 90° (d) 60°

- 69.** What is the value of $\angle BAC - \angle OBC$ if A is the point on the minor arc?

- (a) 90° (b) 180° (c) 130° (d) 50°

PREVIOUS YEARS' QUESTIONS

- 70.** A circular ring with centre O is kept in the vertical position by two weightless thin strings TP and TQ attached to the ring at P and Q . The line OT meets the ring at E where as a tangential string at E meets TP and TQ at A and B , respectively. If the radius of the ring is 5 cm and $OT = 13$ cm, then what is the length of AB ? **2012 I**

- (a) $10/3$ cm (b) $20/3$ cm (c) 10 cm (d) $40/3$ cm

- 71.** The locus of the mid-points of all equal chords in a circle is **2012 I**

- (a) The circumference of the circle concentric with the given circle and having radius equal to the chords
 (b) The circumference of the circle concentric with the given circle and having radius equal to the distance of the chords from the centre
 (c) The circumference of the circle concentric with the given circle and having radius equal to half of the radius of the given circle
 (d) The circumference of the circle concentric with the given circle and having radius equal to half of the distance of the chords from the centre

- 72.** Consider the following statements

- I. The locus of points which are equidistant from two parallel lines is a line parallel to both of them and drawn mid way between them.
 II. The perpendicular distances of any point on this locus line from two original parallel lines are equal. Further, no point outside this locus line has this property.

Which of the statement(s) given above is/are correct? **2012 II**

- (a) Only I (b) Only II
 (c) Both I and II (d) Neither I nor II

- 73.** Consider a circle with centre at O and radius r . Points A and B lie on its circumference and a point M lies outside of it such that M, A and O lie on the same straight line. Then, the ratio of MA to MB is **2013 I**

- (a) equal to 1 (b) equal to r
 (c) greater than 1 (d) less than 1

74. Consider the following statements :

- I. The perpendicular bisector of a chord of a circle does not pass through the centre of the circle.
 II. The angle in a semi-circle is a right angle.

Which of the statement(s) given above is/are correct? **2013 II**

- (a) Only I (b) Only II
 (c) Both I and II (d) Neither I nor II

75. The diameter of a circle with centre at C is 50 cm. CP is a radial segment of the circle. AB is a chord perpendicular to CP and passes through P . CP produced intersects the circle at D . If $DP = 18$ cm, then what is the length of AB ?

2013 II

- (a) 24 cm (b) 32 cm
 (c) 40 cm (d) 48 cm

76. Consider the following statements in respect of two chords XY and ZT of a circle intersecting at P .

I. $PX \cdot PY = PZ \cdot PT$

II. PXZ and PTY are similar triangles.

Which of the statement(s) given above is/are correct? **2013 II**

- (a) Only I (b) Only II
 (c) Both I and II (d) Neither I nor II

77. In a $\triangle ABC$, $AB = BC = CA$. The ratio of the radius of the circumcircle to that of the incircle is

- (a) 2 : 1 (b) 3 : 1 **2014 I**
 (c) 3 : 2 (d) None of these

78. AB and CD are two chords of a circle meeting externally at P . Then, which of the following is/are correct?

I. $PA \times PD = PC \times PB$

II. $\triangle PAC$ and $\triangle PDB$ are similar.

Select the correct answer using the codes given below **2014 I**

- (a) Only I (b) Only II
 (c) Both I and II (d) Neither I nor II

79. For a triangle, the radius of the circumcircle is double the radius of the inscribed circle, then which one of the following is correct? **2014 II**

- (a) The triangle is a right angled
 (b) The triangle is an isosceles
 (c) The triangle is an equilateral
 (d) None of the above

80. If the chord of an arc of a circle is of length x , the height of the arc is y and the radius of the circle is z . Then, which one of the following is correct? **2014 II**

- (a) $y(2z - y) = x^2$ (b) $y(2z - y) = 4x^2$
 (c) $2y(2z - y) = x^2$ (d) $4y(2z - y) = x^2$

81. If the angle between the radii of a circle is 130° , then the angle between the tangents at the ends of the radii is **2015 I**

- (a) 90° (b) 70° (c) 50° (d) 40°

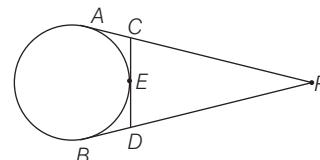
82. Out of two concentric circles, the diameter of the outer circle is 26 cm and the chord MN of length 24 cm is tangent to the inner circle. The radius of the inner circle is **2015 I**

- (a) 5 cm (b) 6 cm (c) 8 cm (d) 10 cm

83. AD is the diameter of a circle and AB is a chord. If $AD = 34$ cm, $AB = 30$ cm, then the distance of AB from the centre of the circle is **2015 I**

- (a) 17 cm (b) 15 cm (c) 13 cm (d) 8 cm

84. From an external point P , tangents PA and PB are drawn to the circle as shown in the above figure.



CD is the tangent to the circle at E . If $AP = 16$ cm, then the perimeter of the $\triangle PCD$ is equal to **2015 II**

- (a) 24 cm (b) 28 cm (c) 30 cm (d) 32 cm

85. Chord CD intersects the diameter AB of a circle at right angle at a point P in the ratio 1:2. If diameter of circle is d , then CD is equal to **2015 II**

- (a) $\frac{\sqrt{2}d}{3}$ (b) $\frac{2d}{3}$ (c) $\frac{2\sqrt{2}d}{3}$ (d) $\frac{2\sqrt{3}d}{3}$

86. The diameter of a wheel that makes 452 revolutions to move 2 km and 26 dm is equal to **2015 II**

- (a) $1\frac{9}{22}$ m (b) $1\frac{13}{22}$ m (c) $2\frac{5}{11}$ m (d) $2\frac{7}{11}$ m

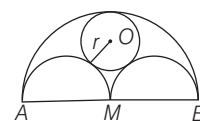
87. A boy is cycling such that the wheels of the cycle are making 140 revolutions per minute. If the radius of the wheel is 30 cm, then the speed of the cycle is **2015 II**

- (a) 15.5 km/h (b) 15.84 km/h (c) 16 km/h (d) 16.36 km/h

88. The two adjacent sides of a cyclic quadrilateral are 2 cm and 5 cm and the angle between them is 60° . If the third side is 3 cm, then the fourth side is of length **2015 II**

- (a) 2 cm (b) 3 cm (c) 4 cm (d) 5 cm

89. AB is a line segment of length $2a$, with M as mid-point. Semi-circles are drawn on one side with AM , MB and AB as diameter as shown in the above figure. A circle with centre O and radius r is drawn such that this circle touches all the three semi-circles. The value of r , is **2015 II**



- (a) $\frac{2a}{3}$ (b) $\frac{a}{2}$ (c) $\frac{a}{3}$ (d) $\frac{a}{4}$